Experiment No. 2

Signal Operations and Transformations

OBJECTIVE:

(i) To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

(ii) To perform time shifting and time reversal of continuous-time and discrete-time signals and analyze the resulting transformations.

(iii) To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics.

(iv) To visualize and compare the original and transformed signals using MATLAB and interpret the effects of different signal operations.

(1) To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

Experimental Procedure

1. Generate the base signal $x(t) = \sin(2\pi 5t)$.
2. Apply amplitude scaling using $A = 2$.
3. Apply time compression using $x(2t)$.
4. Apply time expansion using $x(t/2)$.
5. Plot the original and transformed signals using common axes.
6. Measure the amplitude and period of the transformed signals.
7. Compare the measured results with the theoretical scaling relationships.

MATLAB CODE:

Editor

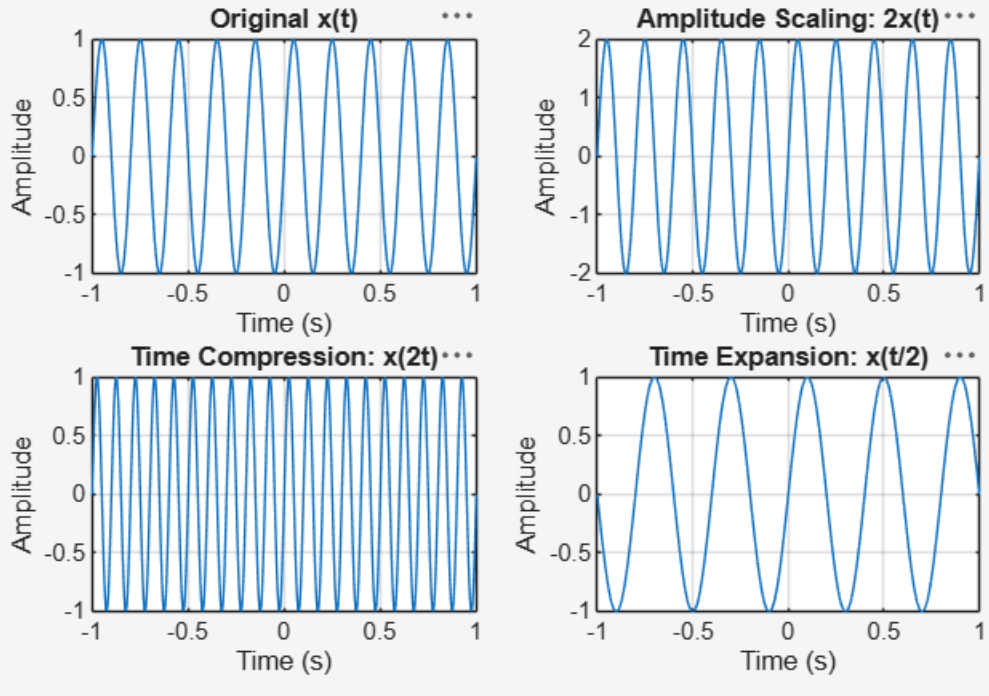
exp_1_a_ss.m | o_exp_5_2_ss.m | exp_1_b_ss.m | exp_2_1_ss.m

/MATLAB Drive/exp_2_1_ss.m

```
% Experiment 2(i): Amplitude and Time Scaling
clc; clear; close all;
t = -1:0.001:1;
x = sin(2*pi*5*t);
yA = 2*x;
yC = sin(2*pi*5*(2*t));
yE = sin(2*pi*5*(t/2));
figure;
subplot(2,2,1); plot(t,x); grid on;
title('Original x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,2); plot(t,yA); grid on;
title('Amplitude Scaling: 2x(t)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,3); plot(t,yC); grid on;
title('Time Compression: x(2t)'); xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,2,4); plot(t,yE); grid on;
title('Time Expansion: x(t/2)'); xlabel('Time (s)'); ylabel('Amplitude');
```

Figures

Figure 1



(3) To perform addition and multiplication of two discrete-time sequences and analyze

the resulting signal characteristics

Experimental Procedure

1. Define the sample index n.

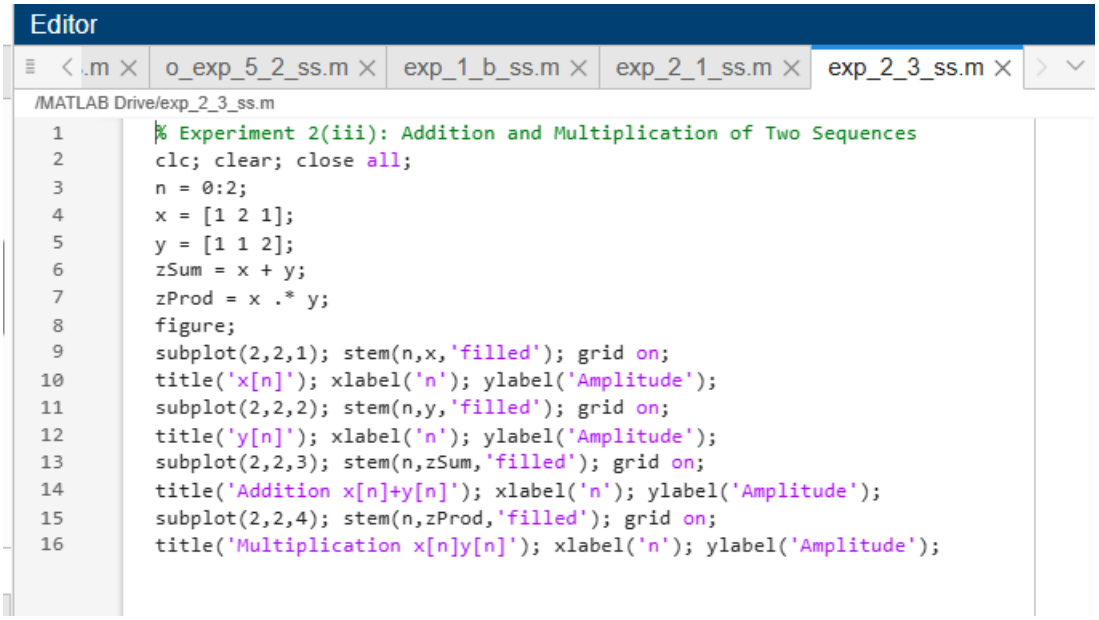
2. Enter x[n] and y[n] with aligned indices.

3. Compute the sum sequence.

4. Compute the product sequence using pointwise multiplication.

5. Plot the input and resulting sequences using stem plots.

6. Compare the calculated samples with manual/theoretical results.



Figures

Figure 1

